

ReLU controls signal and gradient flow

$$a = \text{ReLU}(z) = \max(0, z)$$



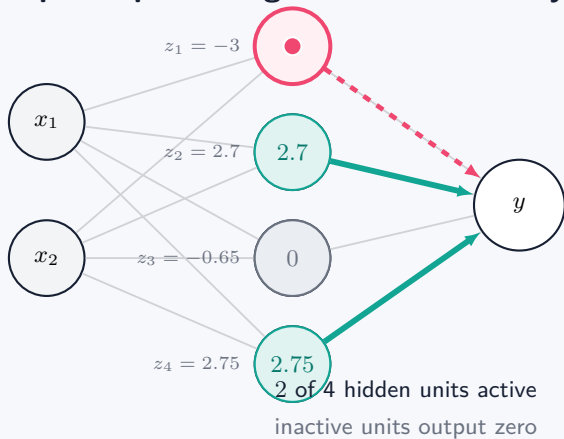
$$\text{ReLU}(z) = \begin{cases} 0, & z \leq 0 \\ z, & z > 0 \end{cases}$$

$$\text{ReLU}'(z) = \begin{cases} 0, & z < 0 \\ 1, & z > 0 \end{cases}$$

$$z = -3 \quad a = 0 \quad \text{ReLU}'(z) = 0$$

BACKPROPAGATION

Input-dependent gates in a hidden layer



$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \text{ReLU}'(z) = \frac{\partial L}{\partial a} \cdot 0 \quad \text{gradient stops}$$